

Lab Worksheet 13  
The Network Layer - IPV4 Addressing

**This Week's Objectives**

1. Be able to configure multiple netkit labs.
  2. Be able to describe how IP addresses are used to route datagrams.
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**Expected Outcomes**

After the laboratory session, students should be able to:

- configure an individual Linux host via ip, /etc/network/interfaces and ifup.
  - configure several hosts to be on the same subnet.
  - explain how netmasks / VLSM and CIDR notation define the portion of the IP address that identifies the network
    - and by implication, how they also define the portion of the address that identifies unique hosts on a network
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**Workplan**

*These lab worksheets assume that you will make accurate notes of everything that you do for later consideration and revision. You should use your lab diary to note down items such as commands used, the outcome of each command, and answers to questions posed in these instructions.*

*All lab machines have CherryTree Notes software installed, which facilitates organised note-taking. Alternatively, you could buy a hardback A4 laboratory notebook.*

*NB this is a complex lab – please read each instruction carefully – if you miss out any steps, then don't expect to achieve the correct results! You will need to have completed and thoroughly understood all the steps before the next lab session.*

*Remember that there is a lab work wiki available to you on BB. Please use this to share and communicate with your colleagues.*

**Make sure you have completed all the steps in this worksheet before the next lab session.**

## Lab Worksheet 13

### The Network Layer - IPV4 Addressing

#### Starting assumptions

1. You have completed all the work in previous lab sheets
  - You have used Netkit *lstart* to launch a coordinated group of several virtual machines. Specifically, you are aware of how this group is coordinated via the *lab.conf* file.
  - You know how to shut down virtual machines using *lhalt* and *lcrash*.
  - You have used *ip* and *ping* within the virtual machines.
  - You have saved captured network traffic from a virtual machine using *tcpdump*.
  - You have viewed a packet capture file in the real Linux host using *wireshark*.

## Activities

#### Reminder

- You need to make accurate notes of what you do.
- You will complete unfinished activities before the next timetabled lab session.
- You will resolve what you do not understand by conducting your own careful (and ethically sound) experimentation and / or further reading.

#### IPV4 Addressing

(IPv4) is the most commonly used version of the Internet Protocol (IP). Each device attached to a network will have a unique IP address of the format 999.999.999.999. Part of the address is the network identifier, and the remainder locates a specific host on that network. The CIDR number (often seen as a /99 at the end of the IP address) tells a network device where the divide between the network and host addresses falls.

Read some background about this topic [here](#) and [here](#).

#### Getting Started

2. Create a directory for lab 13's work. Use a command similar to this:

```
cd ~  
mkdir -p ctec1704/lab_13
```

3. Download the zipped archive *netkit\_lab\_13.tar.xz* from Blackboard.
  - This contains the configuration information for 15 virtual machines.

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4. Extract the contents of the archive file, and make sure that all the files and directories for the lab are in the new *lab\_13* directory (not a sub-directory).

```
.../lab_13:~$ tar -xvf netkit_lab_13.tar.xz
```

5. This netkit lab represents the configuration of nine hosts and five gateways / routers. Note that these are connected via *hubs* rather than *switches* (what is the difference)?

Connectivity is as the ASCII art diagram in the file *lab.conf* (draw this in your own notes to annotate as the lab progresses).

6. Ensure that you are in the newly created *lab\_13* directory and list its contents.

```
...$ cd ~/ctec1704/lab_13
.../lab_13: ls -l
```

7. You will see several directories (some empty, some with content), one for each virtual machine.
8. Launch the virtual machines using the instruction:

```
.../lab_13$ lstart
```

9. You should see virtual machines start up in quick succession. Use the *vlist* command in the host terminal to list all the open netkit terminals. Arrange the windows on the screen so they are consistent with the network diagram you have created in your notes.

### Look at the configuration

10. The subnets have been labelled W, X, Y, Z and D. For each subnet in turn, look at the network configuration of each host / network card connected.

In particular note the IP address, the netmask and the broadcast address of each. Make sure you know how each machine/NIC got its IP address.

```
root@gwW:~# ip addr show dev eth0
```

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11. Start capturing the traffic from the gateway machine and saving it to your *captures* directory. For example, on subnet W, the gateway machine would be *gwW* so the instruction would be:

```
root@gwW:~# tcpdump -s0 -i eth0 -w /hosthome/ctec1704/captures/ip-dumpW1.pcap
```

12. Ping between two machines in the same subnet. Stop capturing the the traffic (Ctrl-c in the terminal where you started the tcpdump) and analyse it using wireshark.
13. Look up the IP header format in [RFC 791](#). Compare what you find in the RFC document with what you see in the middle panel of the wireshark capture for one particular "ping" of your choice.

### Change the configuration

14. Modify the running virtual machines on subnet W so that all the machines are on a different subnet. Specifically, place them all on the 146.227.150.64/26 subnet.

Why can they not go on the 146.227.150.64/25 subnet?

15. Crash and restart the virtual machines on subnet W to restore them to the original configuration.

```
.../lab_13$ lcrash m1 m2 gwW
```

```
.../lab_13$ lstart m1 m2 gwW
```

16. Add an extra machine to the lab:
- Give it the name *foo*, give it one Ethernet card and put it onto subnet X.
  - Record precisely what changes you made so that the extra machine works with the other machines in the lab.
  - You will find you need to edit *lab.conf*, create a directory and then figure out how to ensure it starts up with the correct IP address.

### Extension activities

17. Give two machines on the same subnet the same IP address. What happens when you try and interact with them?
18. Make one or two configuration changes so things are slightly broken. Challenge a colleague to identify and correct the problem.

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19. Look at the *cidr* spreadsheets (there are both Excel and ODF formats - they were created in ODF so that should be more robust). Explain exactly what the spreadsheets show.
20. Experiment ...